Aim:
Mote-to-PC Serial Communication Simulation
Implement serial communication between a sensor mote and PC to monitor real-time data transmission.
Code:

**Mote.ned:**

```
package pc;

simple Mote
{
    parameters:
        double sendInterval @unit(s);
    gates:
        output out;
}
```

**PC.ned:**

```
package pc;

simple PC
{
    gates:
        input in;
}
```

**SerialNetwork.ned:**

```
package pc;

network SerialNetwork
{
    submodules:
        mote: Mote
        {
            parameters:
                sendInterval = 1s;
        }

        pc: PC;

    connections:
        mote.out --> pc.in;
}
```

Jeetesh Shirganoor

**Mote.cc:**

```
#include <omnetpp.h>

class Mote : public cSimpleModule
{
  private:
    cMessage *timer;

  protected:

    virtual void initialize()
    {
        timer = new cMessage("timer");
        scheduleAt(simTime()+1, timer);
    }

    virtual void handleMessage(cMessage *msg)
    {
        if(msg == timer)
        {
            cMessage *data = new cMessage("Temperature=25C");

            EV << "Mote Sending Data\n";

            send(data, "out");

            scheduleAt(simTime()+1, timer);
        }
    }

    virtual ~Mote()
    {
        cancelAndDelete(timer);
    }
};

Define_Module(Mote);
```

Jeetesh Shirganoor

**PC.cc:**

```
#include <omnetpp.h>

class PC : public cSimpleModule
{
  protected:

    virtual void handleMessage(cMessage *msg)
    {
        EV << "PC Received: "
           << msg->getName()
           << "\n";

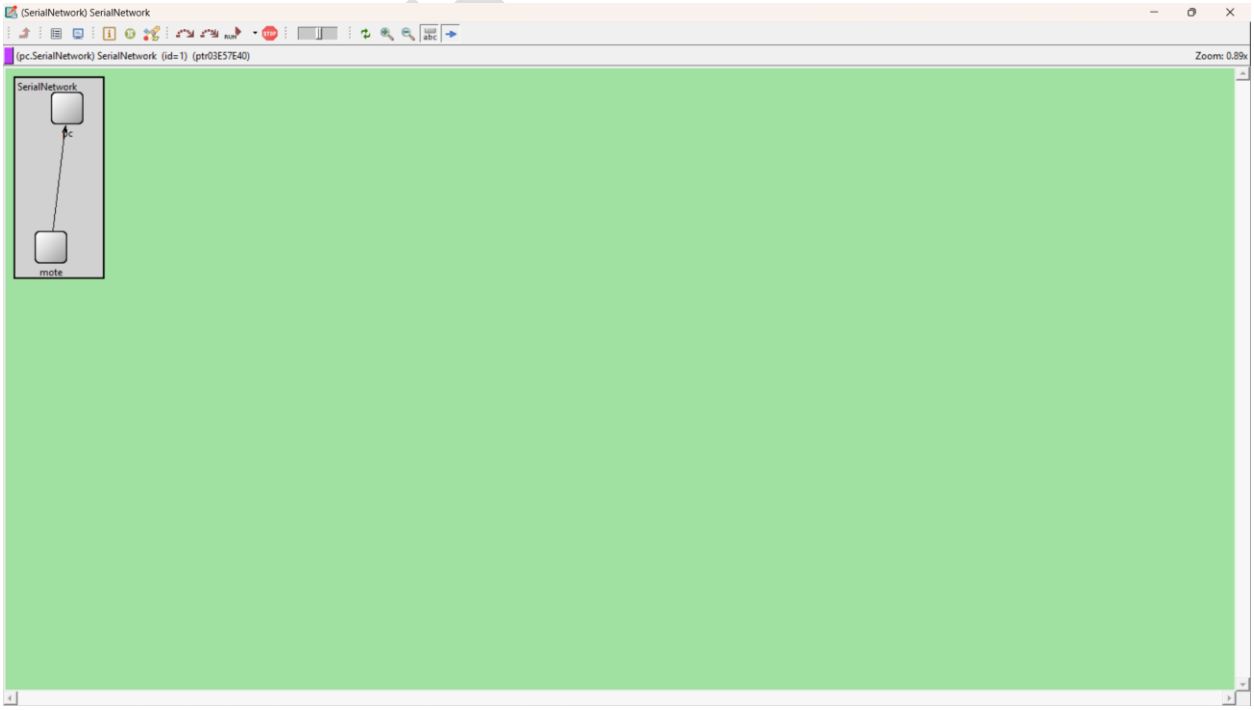
        delete msg;
    }
};
Define_Module(PC);
```

**omnetpp.ini:**

```
[General]
network = pc.SerialNetwork
sim-time-limit = 20s
```

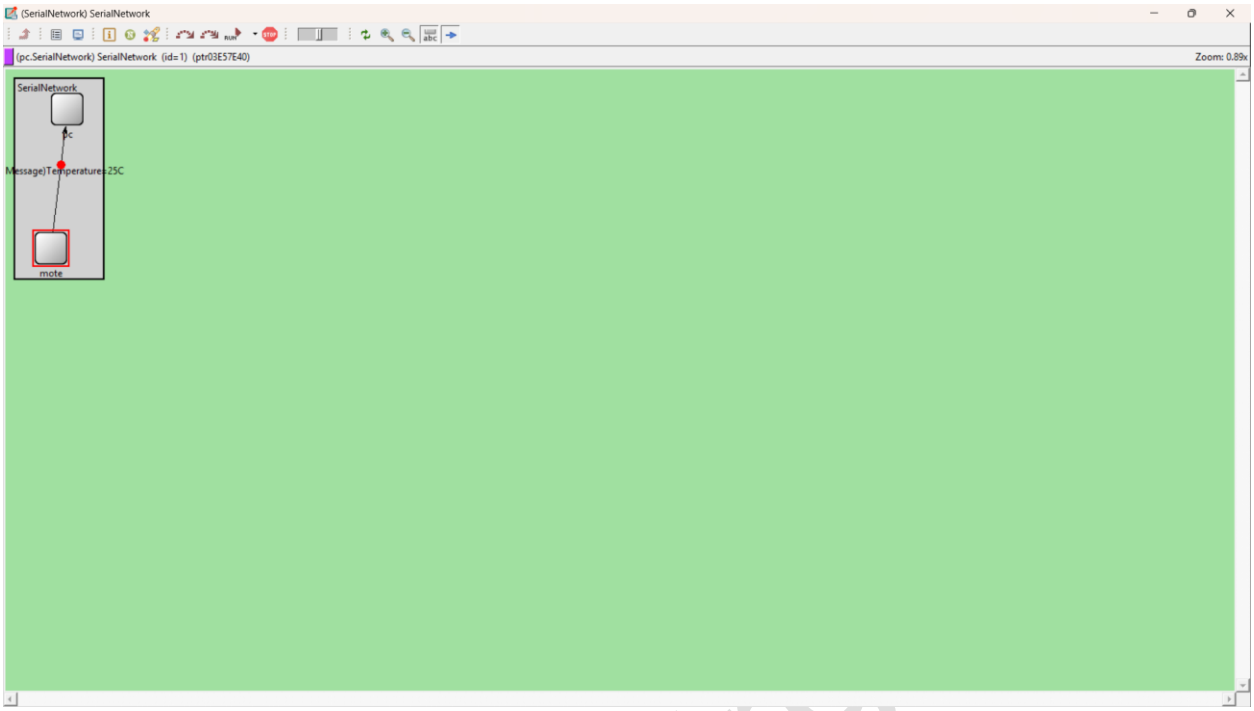
**Output:**

Serial Network Topology



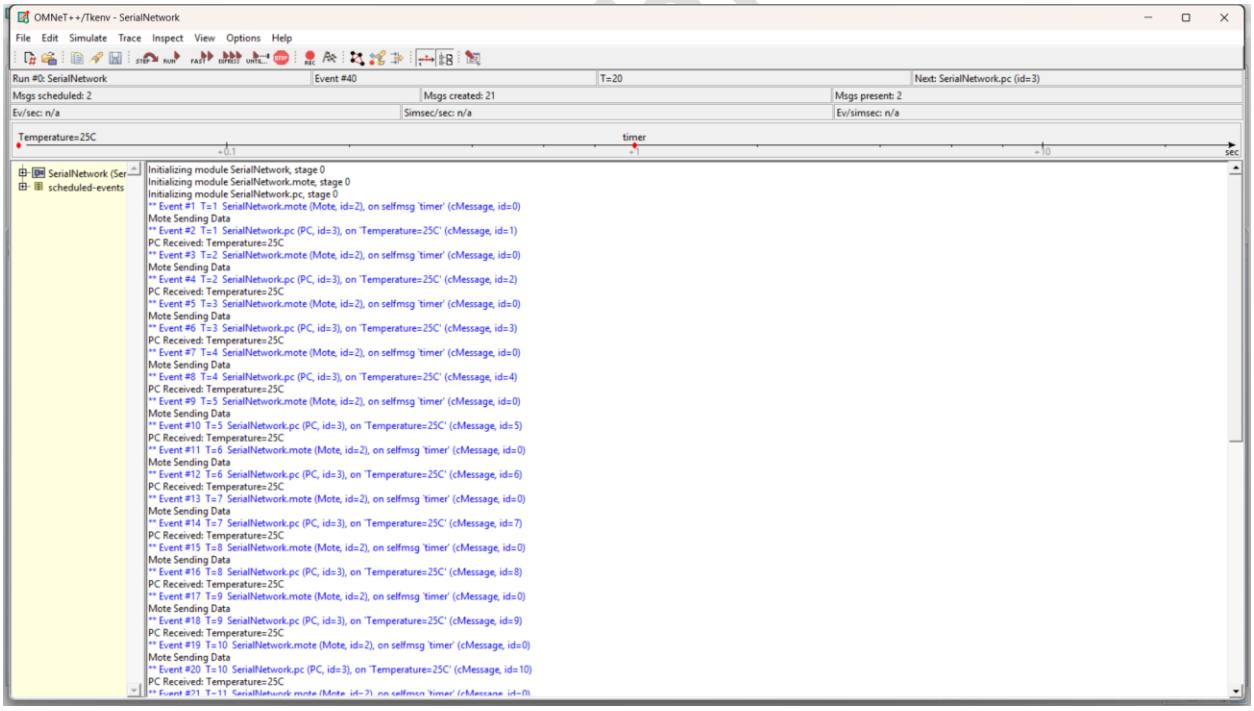

Simulation Execution:

Mvlu College
Wireless sensor network




Mote Sending Data Log

PC Receiving Data Log




## Result

The sensor mote successfully transmitted data to the PC through a serial communication link. The PC received and displayed the transmitted data in real time during the simulation.